

GABLS1: filtered wall drag

GABLS1 filtered rough-wall drag | 23 September 2026. Two new 32³, 12.5 m, WENO9 cases use a 300 s temporally filtered wall state, matched against earlier unfiltered no-closure and scheme-native one-face SLD controls. All use a 400 m cube, seed 123, and nine hours of the same cooling and geostrophic forcing. The fixed 1 m LES median is a model intercomparison reference, not observations.

Filtering improves no-closure near-wall shear but leaves the SLD deficit almost unchanged. In 8–9 h, mean-wind vector shear across 6.25–18.75 m falls from 0.1861 to 0.1541 s⁻¹ without an interior closure, versus 0.1004 s⁻¹ for the fixed 1 m median profiles. With scheme-native SLD it changes only from 0.0522 to 0.0530 s⁻¹. The same ordering appears during 7–8 h. These are adjacent-cell-center gradients, not wall gradients; reference u and v medians were sampled at those heights before differentiation.

In the no-closure pair, filtered wall drag raises final-hour mean u^* from 0.2794 to 0.3019 m/s, makes mean kinematic surface heat flux more negative (-0.01146 to -0.01314 K m/s), and raises peak resolved w^2 from 0.0864 to 0.1013 m²/s². For SLD, u^* rises from 0.2901 to 0.2949 m/s and peak w^2 from 0.05945 to 0.06276 m²/s². Both SLD cases continue to suppress resolved near-wall w^2 and w^3 relative to no closure. A single turbulent realization cannot establish a systematic effect beyond this matched sensitivity test.

Filtering barely changes SLD's first-face mean momentum viscosity (1.2958 to 1.2892 m²/s) or heat diffusivity (1.2642 to 1.2396 m²/s). Filtered SLD's measured WENO u-flux correction is -0.000256 m²/s² versus covariance -0.009609 m²/s², about 2.7%; its heat correction is approximately zero. The small numerical correction does not explain the persistent SLD shear deficit.

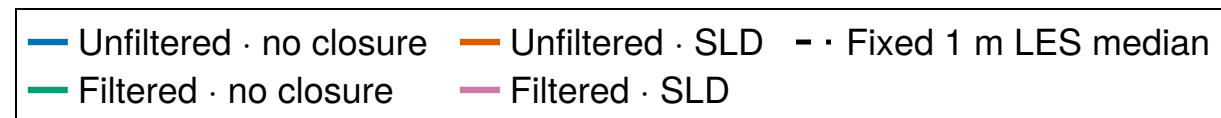
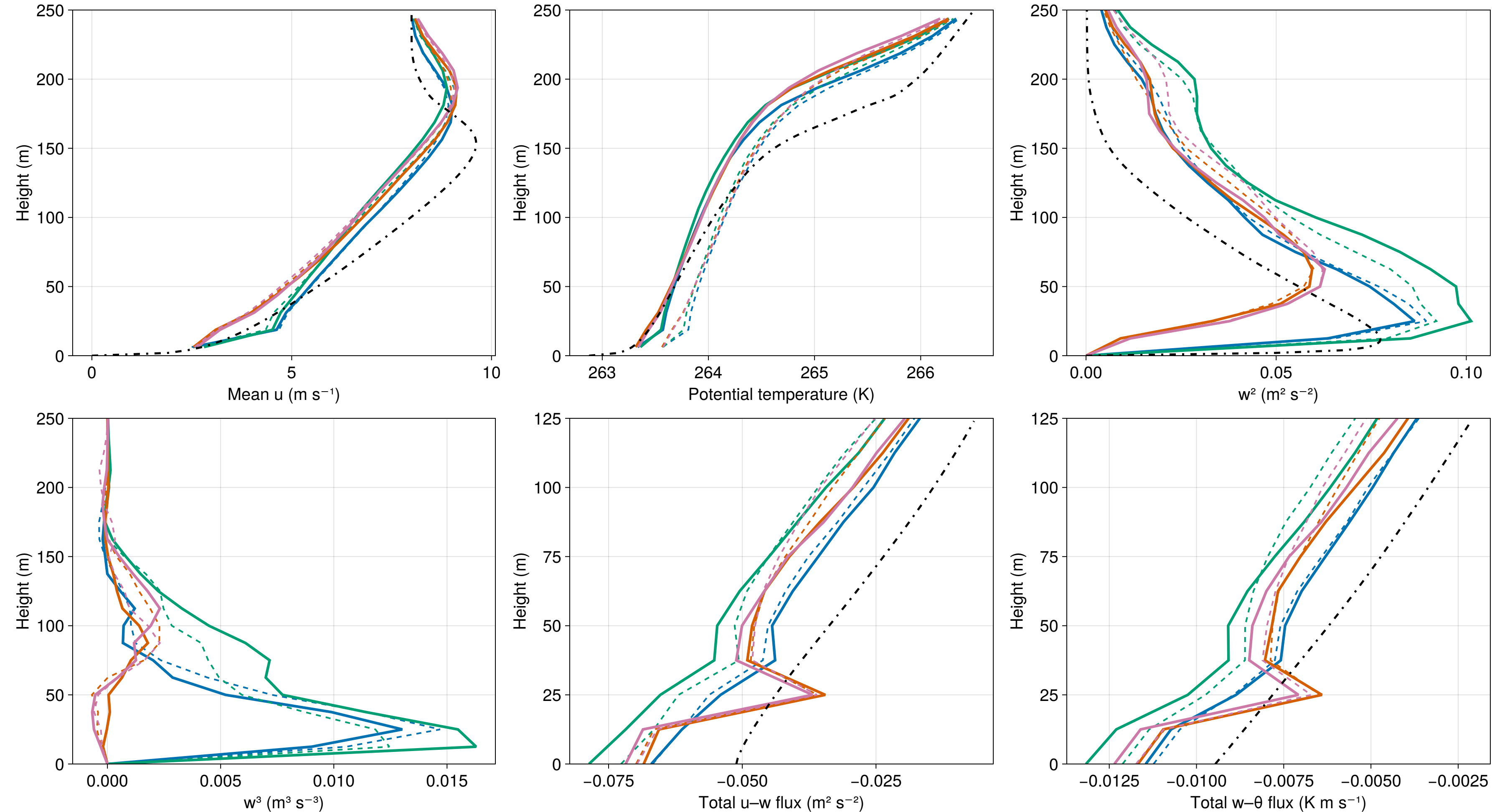
Validation: corrected H100 gate 7536 passed 19/19 no-closure and 21/21 SLD checks. Both science-array 7537 tasks ended with durable exit zero and CASE_DONE at 32400 s.

Immutable source hash manifest

61a5a464b9d100ff2f302ad822972af9ee47e9340a25bbae0ff597890911367e was rechecked; initial theta digest matched across cases. Independent Julia raw audits verified exact 1/19/541/55 initial/profile/series/filter schedules, native 32/33 vertical levels, finite output, evolving filtered state, density-consistent wall fluxes, and both time windows. Both-face reconstructed flux identities passed for SLD. The failed first gate 7534 launched no science and remains historical evidence.

GABLS1: filtering the rough-wall bulk drag

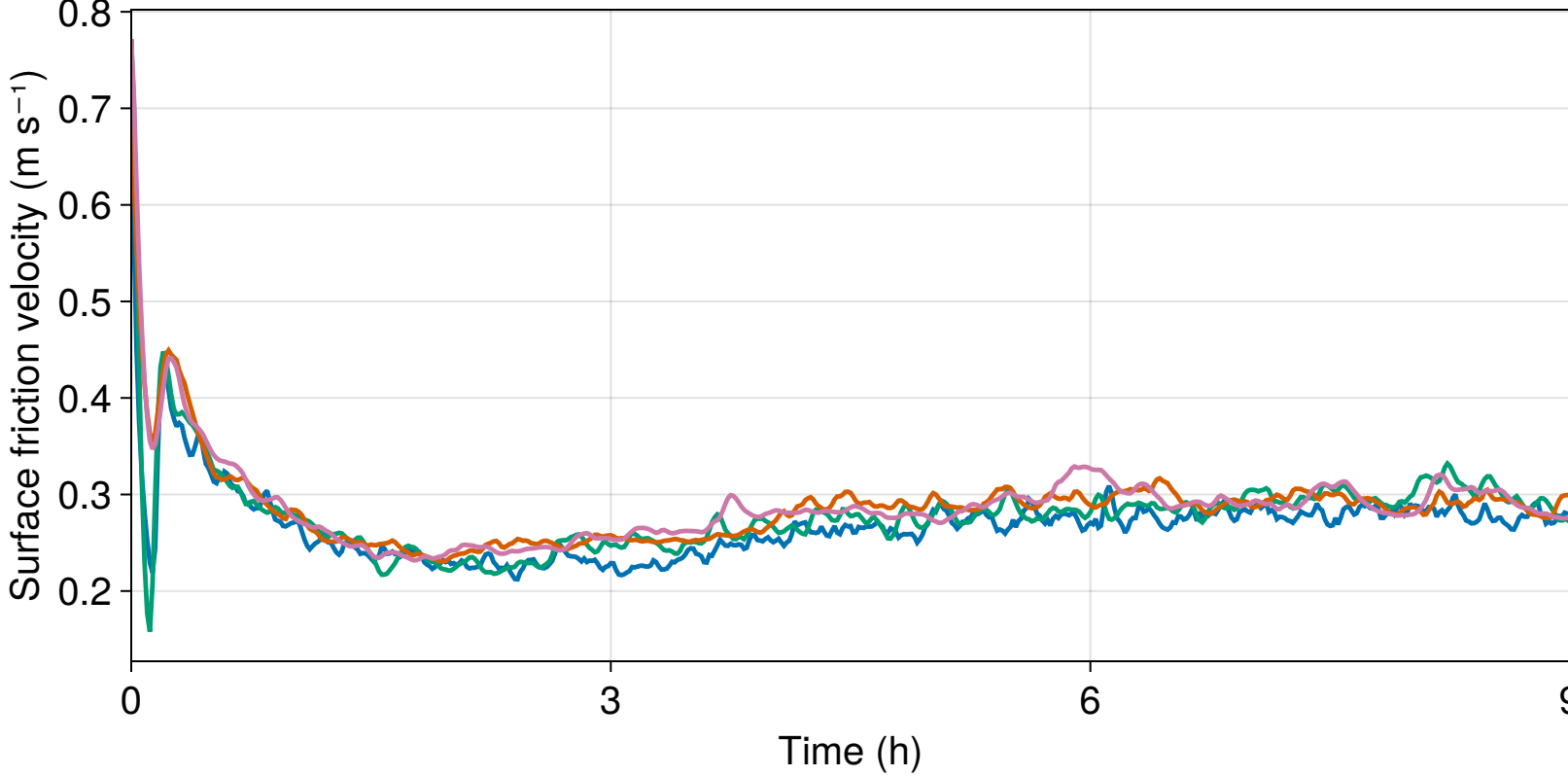
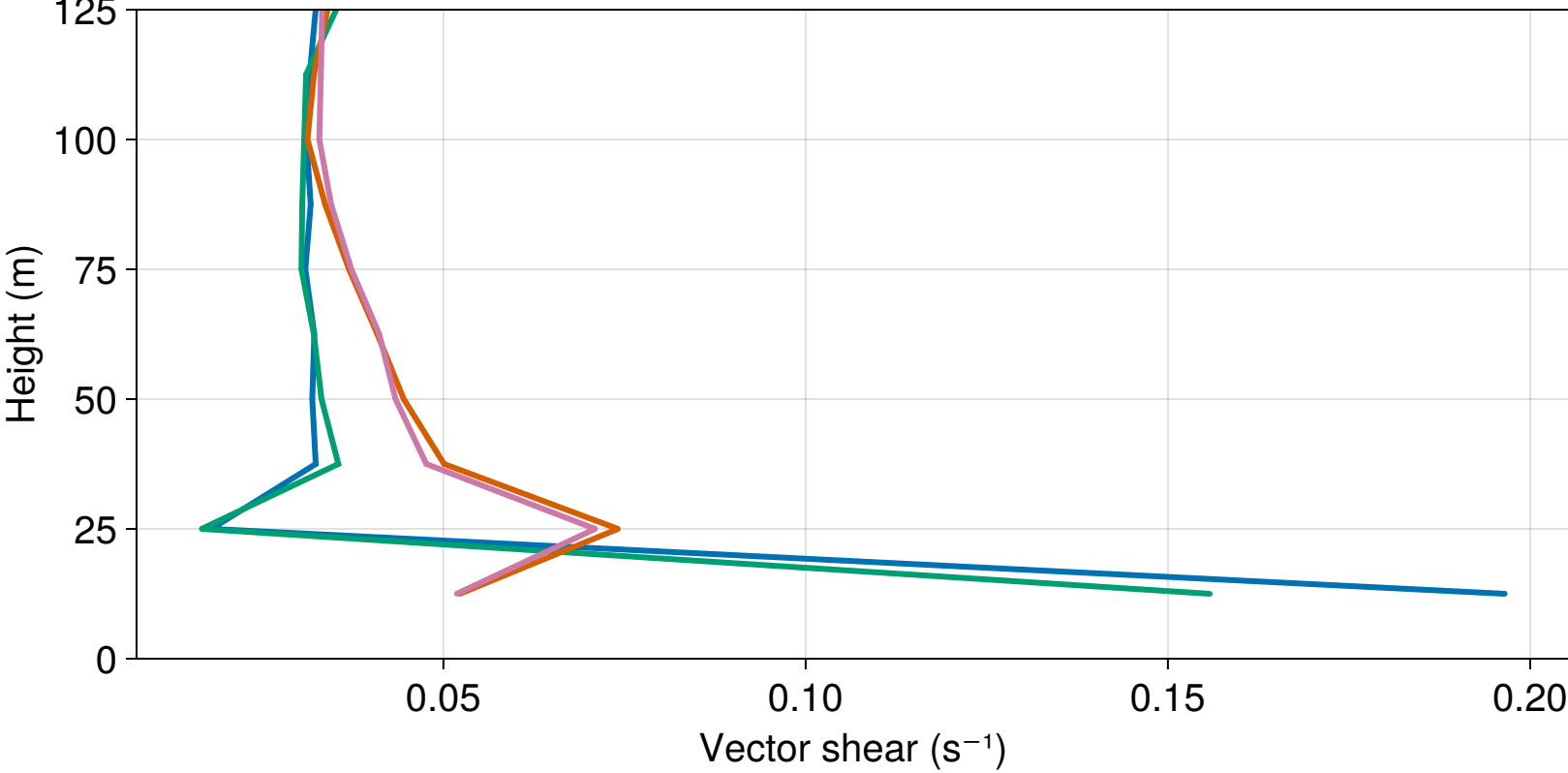
32³ · 12.5 m · WENO9 · 300 s filtered wall state · same seed · dashed 7–8 h, solid 8–9 h



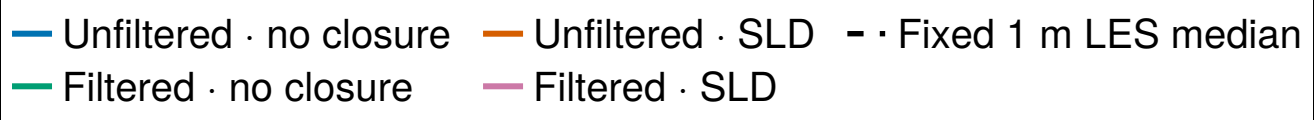
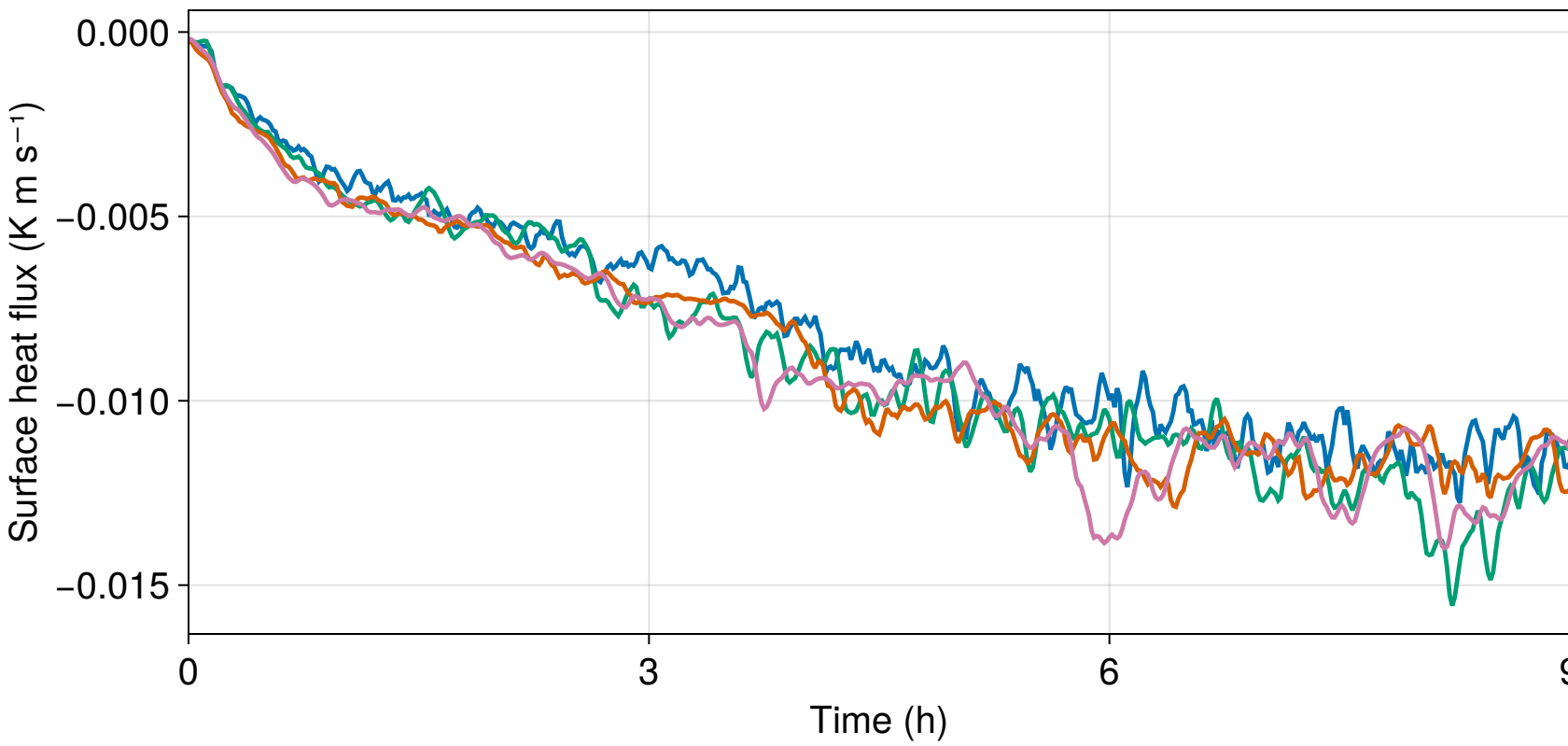
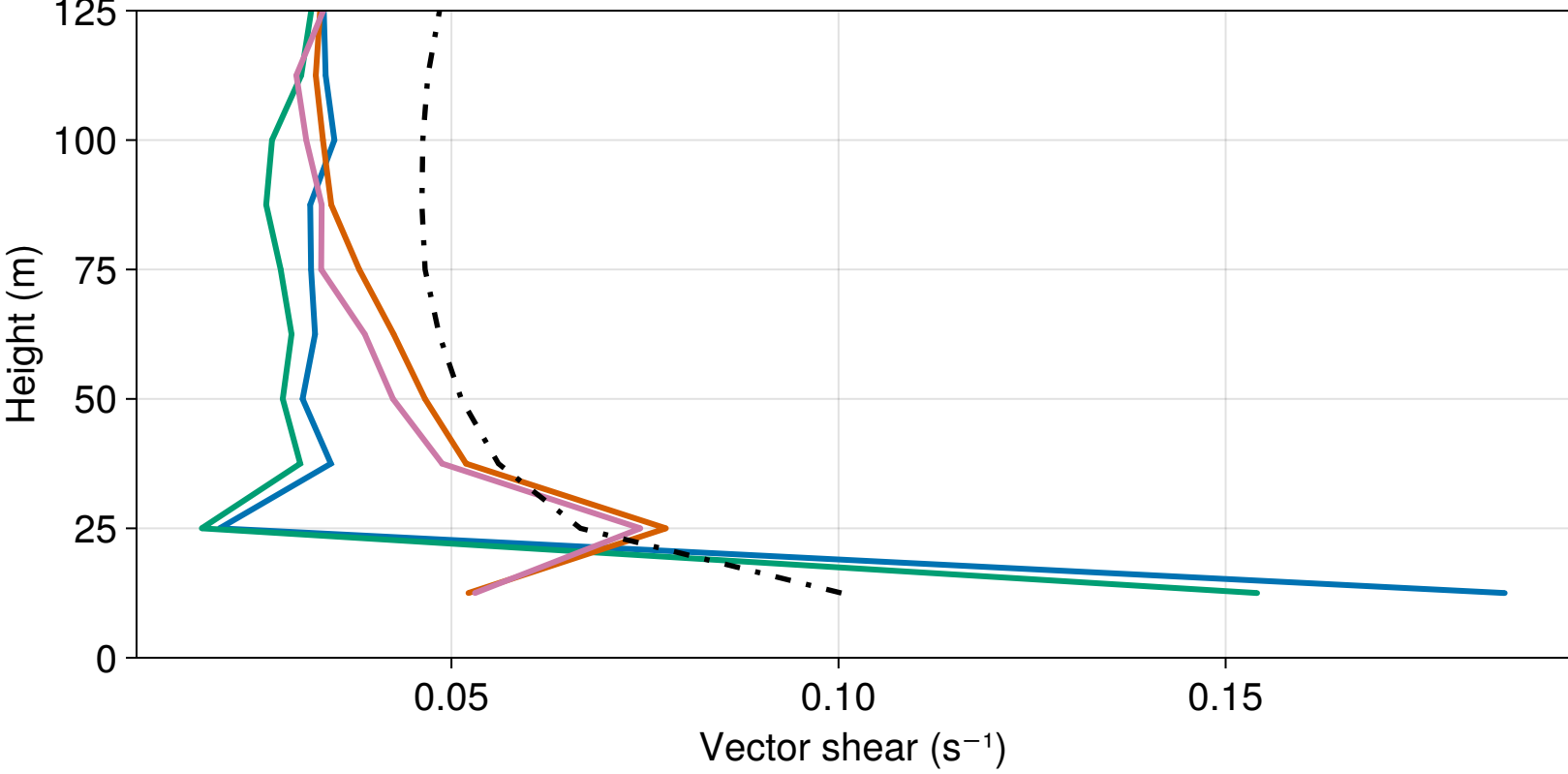
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h



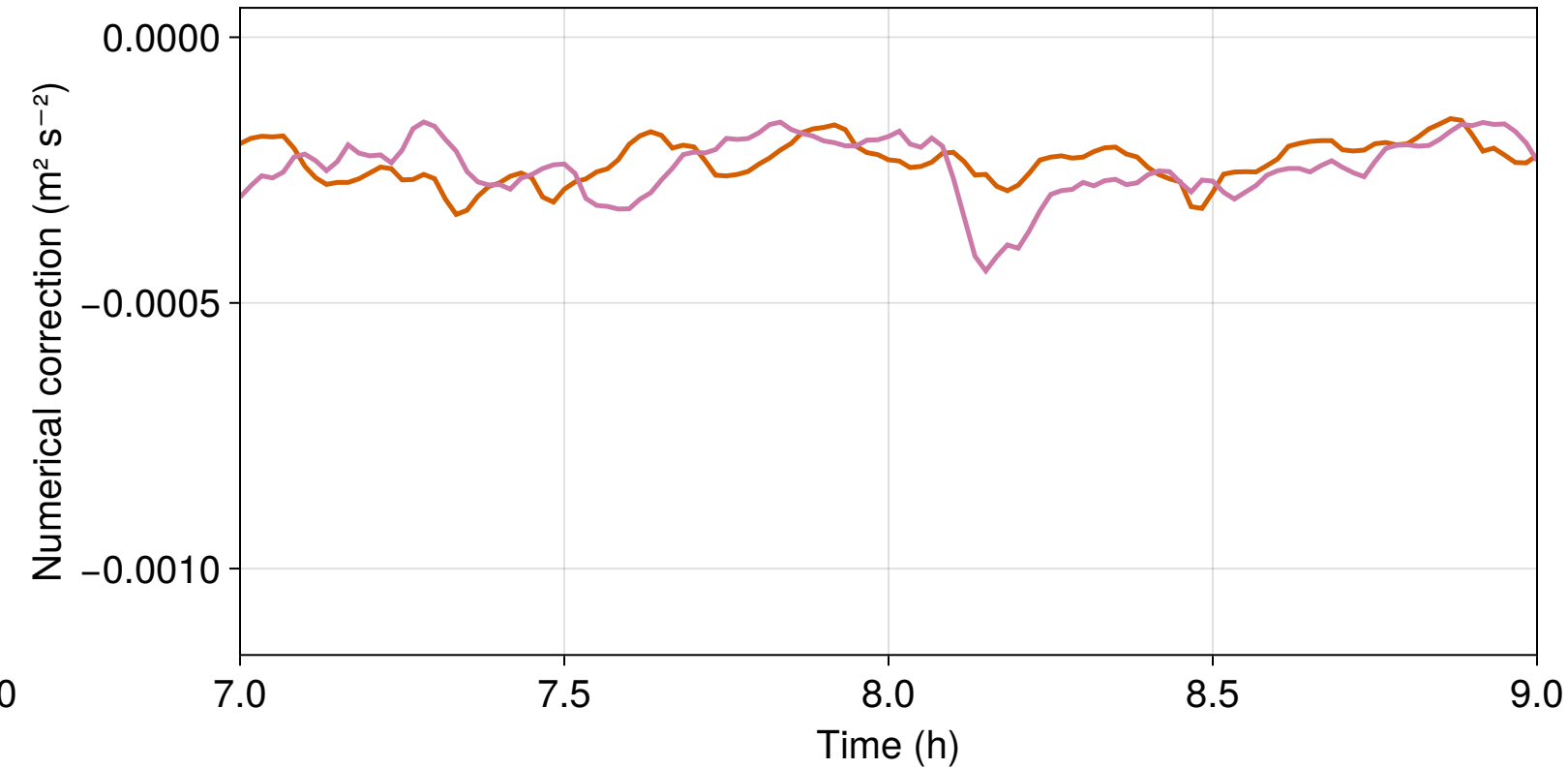
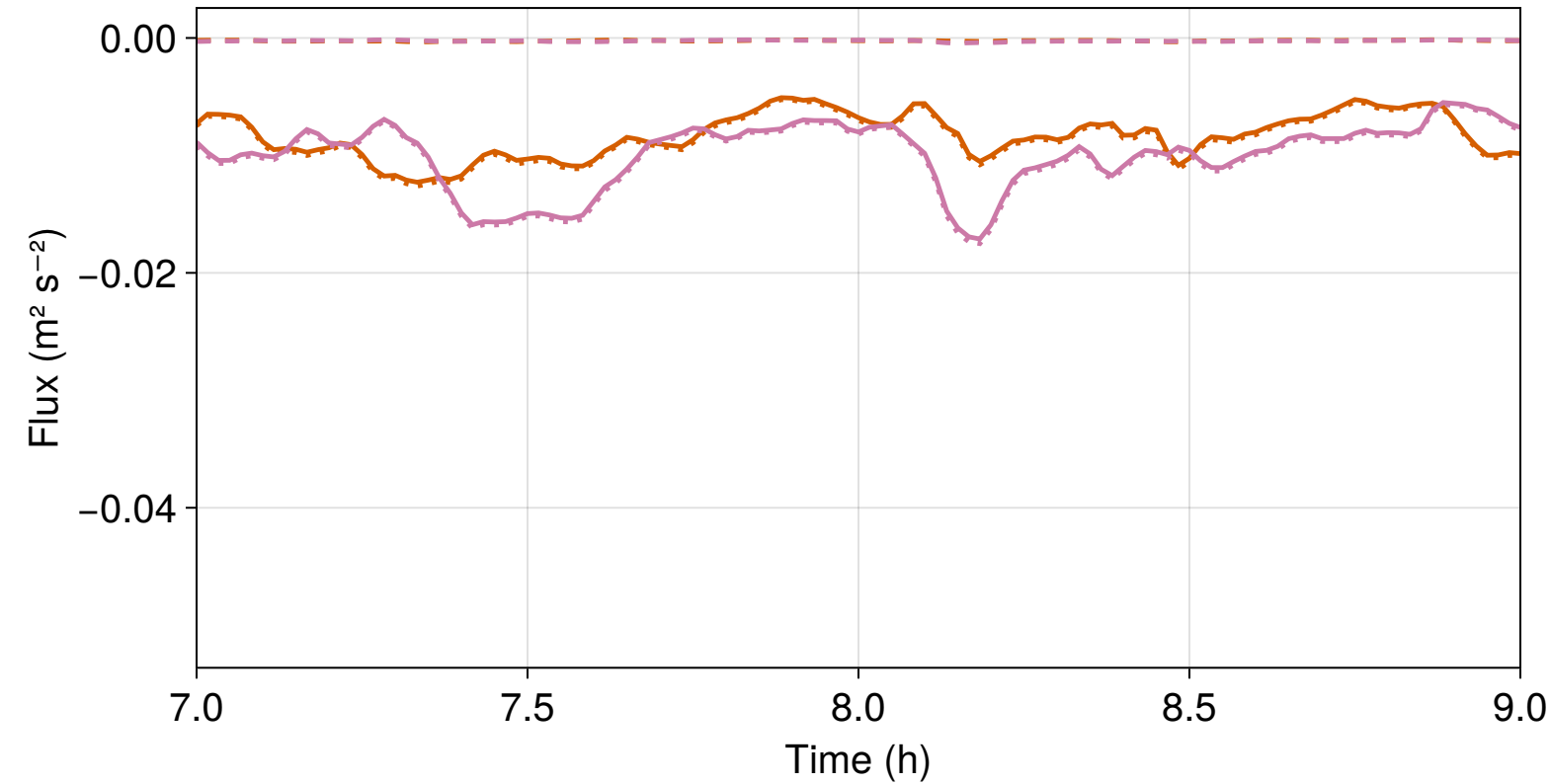
8–9 h



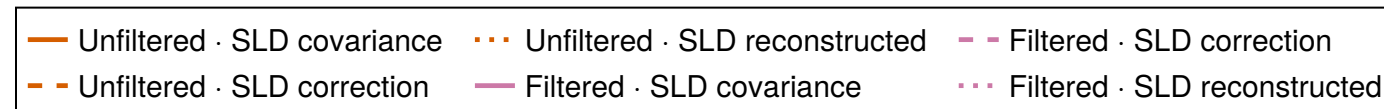
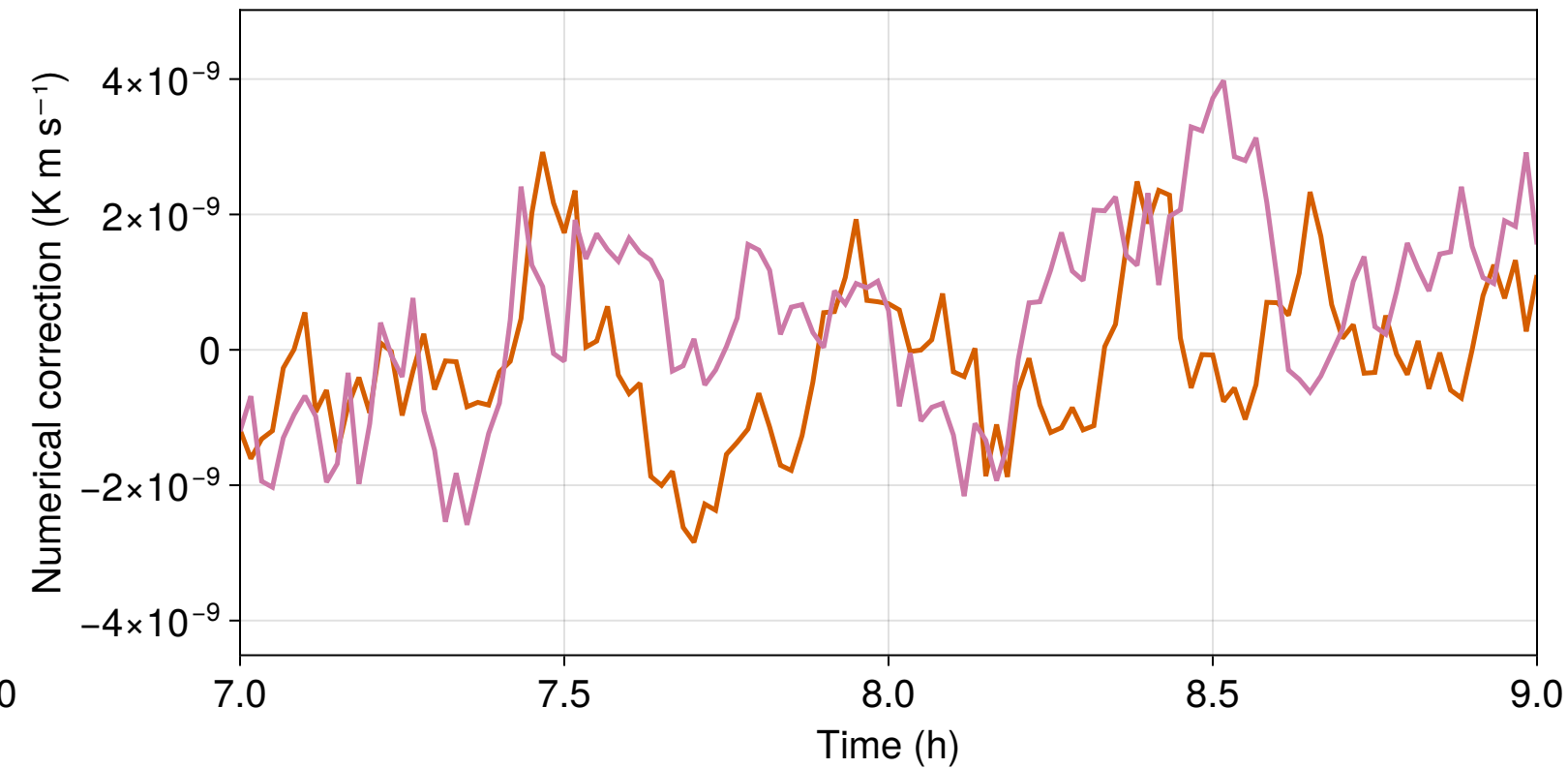
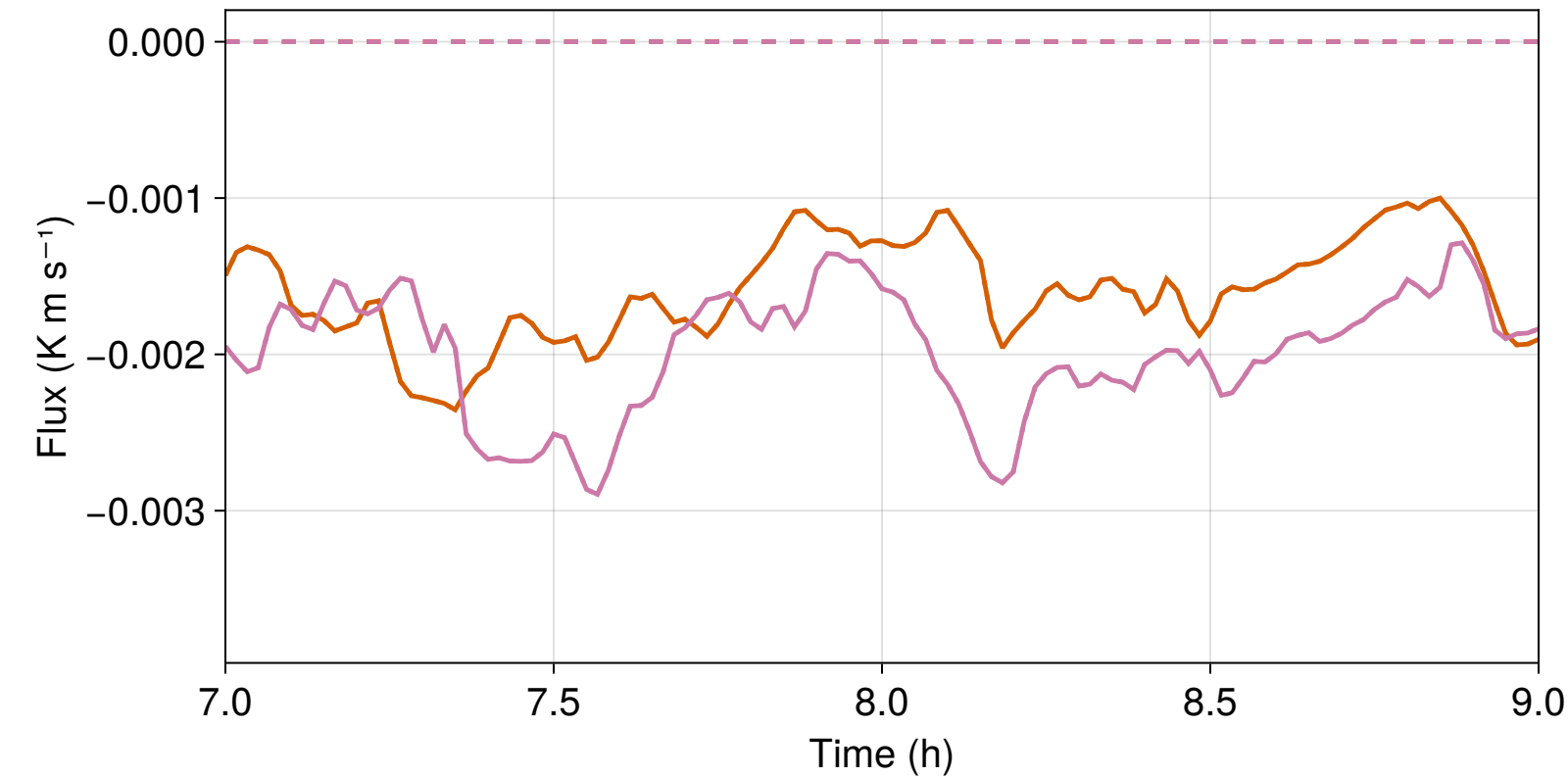
Scheme-native SLD: first-face transport partition

Filtering changes the wall state; reconstructed = covariance + numerical correction

Momentum



Heat



Saved filtered wall state

Horizontal mean of the actual 300 s filtered fields, sampled every 600 s

